

A Shared Neural Prior with Signal-Routed Physics Operators for Cross-Platform Inertial Velocity Estimation

Kaggle team name: thisray
 Ssu-Rui Lee (Team Leader)
 RaydioTek, Taiwan ssr.lee@raydiotek.tw

Abstract— We present an offline system for the TartanIMU Challenge that estimates body-frame velocity from raw six-axis IMU across car, drone, quadruped, and handheld motion. A shared 43.7 M-parameter network provides a velocity prior, and a shared 267-parameter readout combines it with two strapdown proposals using physical residuals and learned corrections. Fixed rules computed from each trajectory activate further rotor-drag or segmented strapdown refinements. Training uses the released training and validation data; inference uses one averaged network weight set and requires the complete trajectory. Our primary submission scores 0.18775 on the organizers’ evaluation over all 89 test trajectories, with AVE 0.13676 m/s and ATE₂₀ 0.59363 m. A sequential ablation for the three-member system reduces the score from 0.27222 for the network alone to 0.18775 for the full pipeline. Aerial motion remains the largest source of error. We report the selection and unsuccessful experiments that informed the final design, including cases where better physical fits or rotation-invariance proxies did not yield better velocity estimates. Transfer to new sensor mountings and storage conventions remains limited.

I. Introduction

The challenge asks for the mean body-frame velocity in each non-overlapping one-second window of a 200 Hz, six-axis IMU sequence. One model must serve car, quadruped, aerial, and handheld motion, which differ in vibration, attitude change, duration, and speed, and the score weights the four platforms equally.

The platforms differ in the constraints available to supplement an IMU measurement. Stationary intervals constrain drift in ground and handheld motion, while rotor drag supplies a velocity-dependent force model for multirotors. A shared regressor must learn across these different motion regimes. The score evaluates the velocities directly and also integrates them with ground-truth orientation, which tests their temporal consistency without requiring the system to estimate orientation or position.

Our system combines a shared long-context velocity prior, a compact readout that mixes it with physical proposals, and fixed signal-based rules that activate further corrections. Every trajectory passes through the same weights and readout. The organizers’ required compliance declaration is in Appendix B.

The design combines temporal velocity regression, as used in learned inertial odometry [1], [2], [3], [4], [5], with strapdown integration constrained by zero-velocity updates (ZUPT) [6] and by rotor-drag models on aerial

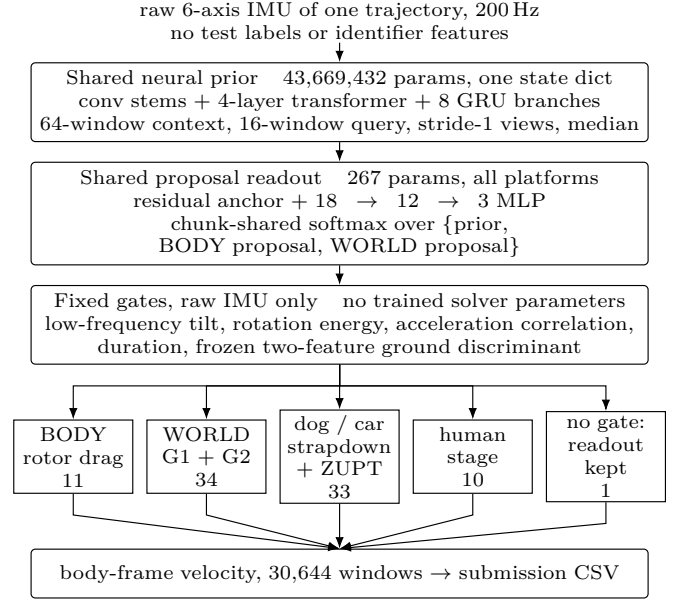


Fig. 1. Inference graph. Every trajectory passes through the shared network and readout. Bold counts refer to the five top-level groups, which are disjoint on this test set. WORLD G2 is a further refinement within the WORLD group. A rejected refinement returns its incoming prediction.

vehicles [7]. Our contribution is the arbitration between the two inside one reproducible inference program: a learned, shared readout for continuous trust, and raw-signal gates for selecting operators. The final network is a weight-space average of related fine-tuned checkpoints [8], not a prediction ensemble. The result is a benchmark-specific offline hybrid, and it is weakest on aerial motion, which accounts for about half of the remaining score.

II. Method

A. Overall Architecture

In Fig. 1, deterministic physics stages follow the shared network and proposal readout. The readout contains the 267 gradient-trained parameters downstream of the network. Stage-specific guards reject detected failures, but incorrect routing can still pass them; Section IV-F examines this limitation.

The backbone encodes each 200-sample window with convolutional stems and a four-layer Transformer. Eight

jointly trained GRU branches process up to 64 windows from the same trajectory and emit the central 16 query windows. Their mean is combined with a shared, confidence-gated low-frequency residual in a discrete cosine transform (DCT) basis. Section II-E describes how these overlapping queries become one prediction per window.

BODY and WORLD name two physical modeling hypotheses. Our released-label diagnostics distinguish streams consistent with sensor/body-frame storage from world-referenced streams with a fixed gyroscope-axis convention; we call these body-stored and world-stored. These are diagnostic categories, not test input fields. The program consumes the supplied six channels unchanged and infers which refinement to apply from their signal statistics. Appendix B gives the storage-law interpretation; Section IV-E examines its association with routing and its implications for transfer.

B. Data Processing

The model consumes raw accelerometer and gyroscope samples only, at the released 200 Hz, with no resampling, gravity alignment, or orientation canonicalization. A window is exactly 200 samples. Training groups windows into 16-window chunks with a 64-window recurrent context at stride one, cycles platforms in the sampler, shifts windows in five-sample phase increments up to 195 samples, and pads trajectories shorter than one chunk rather than dropping them. Context is clipped to the same trajectory, and only valid indexed windows are written at inference.

All members use accelerometer bias jitter (range 0.1 m/s^2) and gyroscope bias jitter (range 0.002 rad/s). The three recipes are TV, no rotation augmentation; A2, body-coordinate tilts of up to 45° ; and SO3, Haar-uniform rotations. A2 and SO3 apply their rotations with probability 0.5 to an allowlist of 147 body-stored trajectories in the released training split, across all four platforms. Final members train on train plus validation, but the rotation allowlist excludes validation trajectories and world-stored streams. The released partition was not resplit.

C. Training Data

Table I lists every dataset used. No external data, synthetic datasets, pseudo-label datasets, or external pretrained weights entered the system; every neural-prior member descends from a warm start we trained on the released training split alone.

D. Training and Optimization

Training has two stages. A warm start is fine-tuned on the released training split only; every deployed member then continues from that same warm start on released train plus validation for a fixed 50 epochs. The optimizer is AdamW, learning rate 2×10^{-5} , weight decay 0.01, batch size 4, 2,048 samples per epoch,

TABLE I
Datasets used for development and training.

Dataset	Type	Traj.	Win.	Usage
Challenge train	Official	395	81,931	training, tuning
Challenge val	Official	80	23,714	final training, readout, selection
Challenge test	Official	89	30,644	inference, scored candidate comparison
External / public	—	0	0	not used
Synthetic / pseudo	—	0	0	not used
Pretrained weights	—	0	0	not used

mixed precision, with an exponential moving average (EMA) of decay 0.999. The loss is an unscheduled weighted sum: windowed AVE (1.0), direct AVE (0.25), body-frame DCT (0.2), uncertainty (0.05), multi-horizon path shape (0.05, horizons 2/4/8/16 windows), per-branch AVE (0.25), branch decorrelation (0.1), and an auxiliary platform-classification head (0.1). These values come from the train+val fine-tuning configuration `df1_branch8_h64_trainval50_gb10_s17.yaml`, released with the team report; A2 and SO3 differ from it only in the rotation-augmentation block and, for the seed-18 members, the seed.

Within a training run there is no checkpoint selection: each member is its fixed terminal epoch-50 EMA weight, and no epoch was chosen by any score. Which recipes entered the pool, which members entered the soup, and which finished candidate was submitted were separate decisions; answer 4 in Appendix B states which of them saw test feedback.

The deployed network is the uniform element-wise average, in a float64 accumulator, of three such members. We call the three-member soup (TV, A2, and SO3 at seed 17) S-C3, and the six-member soup, which adds the same recipes at seed 18, S-C6. Each soup has its own 267-parameter readout, fitted on the 80 released validation labels with seed 17 and 300 AdamW steps under an AVE loss, using that soup’s train-only twin, the same recipes fine-tuned on the training split only, as the input prior, so the readout never saw a prior already trained on the trajectories it was fitted to. Candidate comparison used three-fold out-of-fold readout predictions; the deployed readout is refitted on all 80. Validation therefore remains part of both fitting and selection.

E. Inference

The complete pipeline is one program. Every 16-window query view is evaluated at stride one. Within a view the velocity is the mean of the eight GRU branches plus one shared, confidence-gated low-frequency DCT residual; across views, a window’s prior is the coordinate-wise median of the views covering it (1 to 16 views, 16 for interior windows), a form of temporal test-time augmentation (Appendix B). An interior prior therefore depends on up to 39 past and 39 future windows; the physics stages that follow read the trajectory as a whole,

so the system is an offline, non-causal smoother.

For each 16-window chunk, we fit BODY- and WORLD-frame velocity proposals to the incoming prior using the raw IMU. Eighteen features f summarize proposal residuals, prior motion, IMU statistics, and fit conditioning. The first two are log-transformed RMS proposal corrections relative to the prior. The shared readout uses a fixed residual-based logit anchor and a learned correction:

$$r = \max(\exp(f_{1:2}) - 1, 0), \quad \ell = \text{logsoftmax}(-r/0.10),$$

$$z = \text{clip}((f - c)/s, -8, 8),$$

$$\alpha = \text{softmax}([0, \ell_1, \ell_2] + W_2 \tanh(W_1 z + b_1) + b_2),$$

$$v_t = \sum_{j=1}^3 \alpha_j v_{j,t}.$$

Here c, s are stored feature-standardization statistics, the MLP hidden width is 12, and v_1, v_2, v_3 are the neural prior and the BODY and WORLD proposals. One mixture vector is shared by the windows of a chunk; the candidate velocities vary per window. The residuals measure agreement with the prior, not ground-truth error. Fixed gates then activate the following corrections in order; WORLD has two stages, G1 and G2:

- BODY fits a whole-trajectory rotor-drag least-squares smoother at the deployed coefficient 0.30. When the network-free solution averages at least 2 m/s and at least $1.4\times$ the prior’s mean speed, the incoming prior is first rescaled by that ratio.
- WORLD G1 fits an attitude-conditioned world-frame proposal and switches to pure physics only when its spectral misfit statistic exceeds a frozen bound.
- WORLD G2 integrates the accelerometer once over the whole trajectory, fits the integrated velocity independently in consecutive 20-window segments for the unknowns (v_0, b_a) , fits the otherwise unobservable heading gauge to the incoming G1 estimate, and blends 0.5/0.5 with G1. Its frozen selector, derived on released training data, accepts 20 of the 34 WORLD-gated test trajectories.
- dog and human share one 30s segmented strap-down solver with zero-velocity constraints, blended 0.5/0.5 with the incoming prior. car uses a 60s segmented hybrid solver without that outer blend. Within the ground gate, a frozen linear discriminant over two rotation-invariant features separates car from dog; its three coefficients, including the intercept, were fitted once on released car and dog trajectories. The physics solvers themselves hold no gradient-trained parameters.

The solvers fit per-trajectory variables such as initial velocity, bias, and attitude; coefficients, blend weights, and selector thresholds are fixed at inference. Guards are stage-specific: BODY, dog, car and human revert on non-finite output or any window above 30 m/s. The WORLD G1 fused branch uses residual guards; its selected pure-physics branch checks only finite output. G2 uses its

TABLE II

Model complexity and runtime for the primary system. One window is one second of IMU data.

Property	Value
Parameters, network	43,669,432 (266 tensors)
Parameters, shared readout	267
Trained solver weights	0; ground-gate coefficients: 3
Training hardware	GB10
Retained GB10 trainer clocks	5.05 h (warm start, 2 members)
Inference hardware	one GB10; one Tesla T4
GB10 reference run, wall clock	2,016 s; 28% of the 2 h cap
GB10 throughput	15.2 windows/s
GB10 peak GPU / host RSS	601 MiB / 2.01 GiB
T4 wall clock; GPU / RSS	2,067 s; 489 MiB / 1.80 GiB
Online / causal	no; offline smoother

frozen selector. Every rejection returns the incoming prediction. Appendix F lists every such constant with its value and the data it was chosen on.

F. Model Complexity and Runtime

Table II summarizes cost. The system cannot run online; it needs the complete trajectory. The NVIDIA GB10 runs shared a GPU with unrelated jobs, so those times are contended rather than best-case; both tested environments stay well inside the two-hour and 16 GB limits.

No total training cost is claimed. Retained GB10 clocks cover the warm start and two members; one member’s clock and earlier ancestry were not retained. The three seed-18 members of S-C6 add 8.13 T4-hours of Kaggle session time, including setup and export, which is not directly comparable to trainer-only time.

III. Development Experiments

Official results use the organizers’ service over all 89 test trajectories, including Public and Private. Development scores are local: they apply our implementation of the official metric to released validation, normally with train-only twin priors and three-fold out-of-fold readout predictions. These controls separate prior training and readout fitting from the evaluated labels, but repeated design and selection on the same validation set leave it a development set. The final train+validation weights and refitted readout also use those labels.

Training data and candidate selection. Fine-tuning on train plus validation improved the official score from 0.25052 to 0.24684, with AVE and ATE₂₀ improving on all four platforms. It also removed an auxiliary ordinal head, so this comparison does not isolate the data effect. The preregistered soup comparison used the full chain without G2, with train-only twins and out-of-fold readouts: S-C3 scored 0.171247 and S-C6 0.170990. The rule chose the lowest score, preferring fewer members within 0.0001. Adding G2 gave final-chain validation scores 0.169031 and 0.168824. Both candidates were selected as Kaggle finals; on Public and on the official all-89 evaluation the order reversed, with S-C3

ahead (Table III). A post-hoc fit $S(n) = A + B/n$ to the three- and six-member soups predicted the four-member score within 2.1×10^{-5} and a further reduction of about 0.00026 beyond six members. This extrapolation concerns uniform soups of the tested recipes, all with the same architecture and parameter count; it is not a network-capacity study.

Aggregation and confidence. Replacing the mean across the eight recurrent branches with a geometric median worsened validation by 0.014760, so the mean was retained. This branch aggregation is distinct from the coordinate median across query views. We also looked for a signal that identifies the best view of a window. On an earlier checkpoint, the predicted error scale correlated with error across windows (Spearman 0.769) but only weakly across views of the same window (0.087). For the deployed architecture, the RMS disagreement among the eight branches correlated with view error within a window at only 0.002. Neither quantity was a useful view selector.

Physical fitting and downstream error. The rotor-drag coefficient was well identified on 48 of 52 released body-stored drone trajectories. Substituting those values into the S-C6 train-only-twin validation chain slightly worsened its score (+0.000153). Even choosing the coefficient with labels over the tested 0.15–0.40 grid improved it by only 0.00066; the score-optimal value was often far from the physical one. Accurate coefficient identification therefore did not improve this hybrid chain.

A related failure appeared in physics test-time adaptation. Across 16 preregistered validation trajectories, the fitting objective fell by more than 99% on every trajectory, yet 11 worsened. All four tested drones improved, while every car and human trajectory worsened. The five improvements occurred on trajectories of at most 60 windows; all those with at least 121 windows worsened. A 1,029-window human trajectory reached $59.6\times$ the prior’s mean speed. Weakly constrained initial velocity and bias offer an explanation, but this sample does not isolate length from platform. A preregistered guard still passed three harmful cases and rejected two beneficial ones. Low fitting residual alone was therefore insufficient to accept these corrections.

The retained BODY rescale also required a speed floor. Lowering it from 2 to 1 m/s increased mean AVE on validation BODY trajectories by 0.0299 m/s and the full validation score by 0.0014. This supports keeping the floor in the tested chain, where enlarging an uncertain velocity vector can increase error.

Rotation and transfer. An exposure-matched, post-freeze check found a 93.3% reduction in the car remount-invariance proxy for the rotation recipe, while its official all-89 score improved by 0.00139. The latter fell below the preregistered 0.002 adoption threshold; that threshold was a decision rule, not a measured noise bound. The rotation recipes remain in the soups for diversity; task-level remount results are in Section IV-E.

TABLE III

Official all-89 results; lower is better. The baseline row gives the organizers’ published full-test components [9]; its score follows from the formula. Rows 2–6 are the S-C3 ladder; its first four rows were scored on 2026-09-17 on intermediate CSVs of that lineage, and their RTE was not retained. The S-C6 final is a separate weight set, not a further rung; both finals were scored on 2026-09-20 after freezing. Bold marks the primary system.

System	Score	ATE20	AVE	RTE@5s
TartanIMU baseline	0.538	1.261	0.461	—
Network only, S-C3	0.27222	0.70125	0.22339	—
+ shared readout	0.24005	0.67233	0.18850	—
+ BODY operator	0.20048	0.61620	0.14882	—
+ WORLD G1, dog, car	0.19503	0.60112	0.14451	—
+ human, G2: S-C3 final	0.18775	0.59363	0.13676	0.68528
S-C6 final	0.18835	0.59437	0.13738	0.68723

Six matched pipeline comparisons further exposed unstable small validation differences: all three gains of at most 0.0013 reversed on Public, while the three of at least 0.0014 kept their sign. The human and G2 stages transferred at about 1.9 times their validation gains. These six observations discouraged selecting finals by tiny validation differences; they do not establish a general noise threshold. Table VII records the unsuccessful configurations and retained costs.

IV. Experiments

A. Official Challenge Performance

Every number for our submissions in Tables III–V comes from the official scoring service, not a local re-implementation, and covers all 89 test trajectories, Public and Private together; the baseline row repeats the organizers’ published full-test components [9]. The score combines normalized macro AVE and macro ATE₂₀ with weights 0.6 and 0.4, and each component weights the four platforms equally:

$$S = 0.6 \frac{\text{AVE}}{0.7356384388} + 0.4 \frac{\text{ATE}_{20}}{3.1160277267}.$$

AVE averages three-dimensional velocity error within each trajectory, then within each platform, then across platforms. ATE₂₀ integrates the predicted body-frame velocities with ground-truth orientation that the pipeline never sees, aligns each segment of about 20 m with a rigid SE(3) transform, and measures the remaining position error. RTE over 5 s displacements [10] is diagnostic only.

Our two selected Kaggle finals share one pipeline and differ only in the averaged network and its readout. The primary system described here is S-C3, submission 56353966 (Public 0.22610); S-C6, submission 56353958 (Public 0.22738), is reported alongside it. CSV MD5s: S-C3 7a3fab85c4da99588f02d90f2e90c2b6; S-C6 5381dadd49140b23558cb18159088391. Private scores and the final ranking were not available when this report was prepared.

TABLE IV

Per-platform results for the primary CSV, official scoring service.
Platform names are the service’s own.

Platform	Traj.	Win.	ATE20	AVE	RTE	Share
Wheeled/Car	18	10,584	.38180	.07493	.26491	14.7%
Handheld/Human	10	11,264	.84750	.06854	.39382	21.9%
Legged/Quad.	16	6,697	.33379	.07565	.35468	13.9%
Aerial/Drone	45	2,099	.81144	.32793	1.72772	49.5%
Average	89	30,644	0.59363	0.13676	0.68528	100%

B. Per-Platform Results

Table IV gives the official breakdown. Share is each platform’s contribution to the equally weighted four-platform mean. The three ground and handheld platforms agree in AVE to within a factor of 1.1 and supply 93.2% of the output windows for about half of the score; aerial supplies 6.8% of the windows and the other half, at more than four times any other platform’s AVE. Human trajectories nonetheless have the largest ATE₂₀. The metric integrates over a fixed 20 m of travel, which spans many more windows for a walking person than for a fast drone, so a small per-window error accumulates into a comparable path error.

C. Per-Sequence Results

The full table is Table V in the appendix. Error is concentrated: the median sequence has AVE 0.155 m/s while five exceed 0.65 m/s and together carry 29.1% of the unweighted sum of per-sequence AVE, with a maximum of 1.909 m/s. This descriptive sum is not the platform-macro score. ATE₂₀ orders the sequences differently; its two largest values, 2.794 and 2.659 m, are not on the sequences with the largest AVE.

D. Ablation Studies

Table III is a sequential ablation of the S-C3 graph. The proposal-and-readout stage supplies -0.03217 of the total -0.08447 change; subsequent physical stages supply -0.05230 , including -0.03957 from BODY. The readout increment includes proposal generation and the fixed residual anchor, not just its 267 learned parameters. Two further ablations went to the Public leaderboard: removing the WORLD stage from the corresponding development chain reverts 31 trajectories to the readout output and costs $+0.0035$ ($0.23826 \rightarrow 0.24177$). The package’s stricter variant scored 0.28152 on Public against 0.22738 for the selected S-C6, but it changes both the BODY detector frequencies and the prior weighting, so that gap does not isolate the routing of the two aerial trajectories it reassigns. Because the ladder adds stages in a fixed order, each row is an increment in that order, interactions included, not an isolated main effect.

E. Generalization Analysis

One state dictionary serves all four platforms, and randomizing the auxiliary platform head leaves every

predicted value on a test batch unchanged, so that head does not reach the velocity output. These experiments do not isolate negative transfer: no matched per-platform expert was trained as a control.

Routing follows the frame convention as well as the platform. Of the 475 released trajectories, 458 activate a stage: BODY on 50 of the 52 body-stored drones, WORLD on 278 of the 285 world-stored drones, quadruped on 41 (40 quadruped, 1 car), car on 56 (55 car, 1 quadruped), and human on all 33 humans. The other 17, nine drones and eight quadrupeds, receive no refinement, and no trajectory triggers two stages. WORLD fires on 278 of 287 world-stored trajectories and on none of the 188 body-stored ones, showing a strong association with the storage classification. Because frame and platform are coupled here, these counts do not isolate their respective effects. Thirty-four of the 89 test trajectories take that path, which makes both world-frame stages conditional on held-out data carrying the same convention. If the gate stops firing and no other stage activates, the readout output is retained. The released data do not determine how routing will behave under new storage conventions.

Sensor remounting. A post-freeze study applied six fixed rotations to body-stored validation streams: 10° and 20° about the sensor x and y axes and about a seeded random axis. Routing is stable at 10° , where one of 41 trajectories changes stage, but not at 20° : on the worst axis, five of the eight BODY-gated trajectories lose BODY and one other trajectory gains it. At network level, the unaugmented train-only warm start degrades by $+148.6\%$ on average and the rotation-augmented train-only twin by $+96.7\%$; both are out of sample. Through the deployed chain, that twin degrades by $+74.5\%$ on average and $+123.4\%$ at worst. This is only partly out of sample, because the readout it feeds was fitted on released validation. The deployed chain itself, evaluated in sample, degrades by $+133.7\%$ on average and $+209.6\%$ at worst. The study was run on the S-C6 family; the two families’ network-level gaps differ by 1.4%, but the chain figures were computed only for S-C6. These degradation figures use a restricted four-platform score, with the drone group limited to body-stored streams, rather than the full 80-trajectory validation score. They also use a 40-trajectory grouping from an earlier frame diagnostic, which classifies one dog trajectory differently from the 41 used for the routing counts.

Rotating an already world-stored stream is not a physical remount: with updated extrinsics the storage transform cancels it, and with stale extrinsics it becomes a time-varying perturbation. Neither case is covered by a constant rotation, and we record that as an open gap.

F. Failure-Mode Analysis

Aerial motion contributes the largest share of the remaining score. On released validation, the oracle ro-

tation from predicted to reference direction had lag-one autocorrelation 0.22. Averaging these corrections over three windows retained about 75% of the per-window oracle gain, while constant corrections did not transfer consistently between trajectory halves. Across five tested priors, mean drone direction error ranged from 20.285° to 21.243° ; their uniform prediction average gave 20.517° , compared with 20.285° for the best member. These results motivate improving window-level direction estimates, without establishing an irreducible error floor.

Routing is the second failure mode, and it is asymmetric. A gate that does not fire only forgoes its stage’s gain. A BODY gate that fires on the wrong trajectory can cause large errors, because the whole prior may be rescaled and the only guard is the 30 m/s bound. Three test trajectories sit within 2.3° of the BODY tilt threshold, and the tightest margins for losing the stage are 0.143° and 0.151° ; the ground discriminant’s tightest test margin is 0.0507, and it already assigns two released trajectories across the car/quadruped boundary. We therefore prioritize BODY false activation; gate margins identify sensitivity but do not measure the probability or expected cost of failure on new recordings.

V. Insights and Lessons Learned

The shared neural prior and proposal readout supported all four challenge platforms, while physical refinements supplied additional constraints in selected motion regimes. The sequential ablation shows the value of this combination in the submitted pipeline. It leaves open how much of the refinement a learned model could absorb; we did not run matched per-platform experts or a controlled network-capacity study.

Aerial motion remains the clearest target for improvement. The tested direction-loss variants and uniform prediction average did not materially improve the strongest prior. We would prioritize window-level direction estimates and their reliability. These findings do not distinguish limited data or representation from ambiguity in the measurements; global heading ambiguity alone does not explain error in body-frame velocity direction.

The physical experiments show why a refinement should be judged within the complete velocity-estimation chain. More accurate drag coefficients did not improve that chain, and a smaller fitting objective could accompany a worse trajectory. The rotation experiments similarly separate a useful invariance proxy from measured task performance. We would retain these downstream comparisons on another dataset and re-evaluate the numerical thresholds there.

Transfer across recording conditions remains unresolved. WORLD refinements depend on the storage convention, and the remount tests show substantial remaining degradation. We would collect aerial recordings spanning slow and fast motion, varied dynamics, and controlled mounting changes, with documented storage

frames and complete recording conditions reserved for evaluation. We did not measure the marginal benefit of additional data per hour, so this is a research priority suggested by the observed failures.

A. Feedback to the Organizers

Recording and frame conventions. Inferring embodiment from IMU is part of the challenge. Our routing analysis also shows a strong association with storage convention. Explicitly documenting those conventions and evaluating new mountings with physically consistent frame transformations would help distinguish transfer across motion regimes from reliance on preprocessing. This is the setting we would most like to see in the next edition.

The per-sequence service exposes a label statistic. The per-trajectory AVE of an all-zero submission equals that trajectory’s mean ground-truth speed, available five times a day without any reverse engineering. A branch of our own team exploited exactly this, as disclosed in answer 4 of Appendix B. Releasing per-sequence diagnostics only after candidate freezing would prevent their use in subsequent selection; coarser diagnostics would reduce the precision of the disclosed statistics.

A published re-execution tolerance would help. Our two measured GPU architectures give identical stage decisions and slightly different values (Appendix B); we did not obtain bit-exact output across them, and a stated tolerance would remove the ambiguity for every team.

One set of instructions. The Rules page, discussion #739576 with the setup page, and the starter repository each gave a different report deadline, and the two report templates ask for different fields.

VI. Conclusion

A shared neural prior, a compact proposal readout, and signal-routed physical refinements form an effective offline system on this benchmark. The S-C3 sequential ablation reduces the official all-89 score from 0.27222 to 0.18775. Remaining aerial error and mounting sensitivity motivate better directional estimates and evaluation under new recording conditions; the present results do not establish transfer beyond the challenge’s data conventions and test-informed development process.

Acknowledgment

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AI-assisted research workflow. This work was carried out through an agentic research workflow designed and directed by Ssu-Rui Lee, who set the objectives and constraints, allocated compute, reviewed the reported results and disclosures, and made the final submission decisions. AI agents (Anthropic Claude and OpenAI Codex, with ChatGPT used for review) performed most of the literature review, implementation, experiments, analysis, and drafting of this report.

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APPENDIX

A. Per-Sequence Results (Full Table)

Table V is the organizer service’s per-sequence output for the same CSV and the same scoring event as Tables III and IV. Sequence identifiers are retained as returned; we do not map them to filenames, platforms, or leaderboard membership.

TABLE V

Per-sequence official results, all 89 test trajectories of the primary submission, transcribed verbatim from the official scoring service’s per-sequence CSV for the same scoring event as Tables III and IV. RTE is reported for analysis only and does not enter the ranking.

Source CSV sha256: 31bde7f0....

Seq. ID	ATE20 (m)	AVE (m/s)	RTE (m)	Seq. ID	ATE20 (m)	AVE (m/s)	RTE (m)	Seq. ID	ATE20 (m)	AVE (m/s)	RTE (m)
1	0.81727	0.16253	1.71848	31	0.70467	0.20086	1.60111	61	0.94985	0.2248	1.00892
2	0.61016	0.20668	1.22078	32	0.48383	0.08383	0.3074	62	0.06149	0.01817	0.0577
3	0.83722	0.26311	1.18413	33	0.31929	0.05037	0.20156	63	0.72046	0.08935	0.46262
4	0.29935	0.17075	0.66457	34	1.17245	0.24436	1.36775	64	0.72583	0.06723	0.33663
5	0.63506	0.23707	1.20249	35	1.2134	0.19637	1.12131	65	0.18266	0.05148	0.15008
6	0.6761	0.08997	0.7576	36	0.96289	0.18095	1.83011	66	1.45101	0.24918	1.7464
7	0.84332	0.19889	0.9795	37	0.69086	0.92118	4.45298	67	0.10934	0.02872	0.11183
8	0.74785	0.04875	0.29398	38	0.60782	0.77976	3.82065	68	0.44367	0.07737	0.17114
9	0.41719	0.11542	0.61133	39	0.40165	0.11679	0.69281	69	0.96253	0.09636	0.55975
10	1.07143	0.08585	0.71448	40	0.65199	0.21861	1.4477	70	0.54444	0.06575	0.3044
11	0.98502	0.47974	2.08604	41	0.97052	0.28051	1.93784	71	0.19544	0.04518	0.16389
12	0.72713	0.42477	1.96229	42	0.41969	0.1208	0.33487	72	0.83932	0.26692	1.24043
13	0.25758	0.03214	0.1193	43	1.01291	0.21858	1.02213	73	0.6521	0.16861	0.49617
14	0.6528	0.21612	1.27008	44	0.26736	0.05481	0.32439	74	0.40612	0.1407	0.64699
15	2.6591	0.13994	0.95719	45	0.09605	0.05975	0.13995	75	0.70157	0.23729	1.32532
16	0.14703	0.03544	0.22026	46	0.3697	0.0432	0.22113	76	0.18869	0.04471	0.1882
17	1.21922	0.65857	2.71719	47	0.99224	0.39667	1.65222	77	0.98276	0.37878	2.00669
18	0.63393	0.23037	1.14795	48	0.24967	0.01917	0.11397	78	0.2588	0.07332	0.17347
19	0.46716	0.15582	0.88148	49	0.71668	0.19445	1.17545	79	0.37055	0.18626	0.80438
20	0.13071	0.05166	0.36425	50	0.56618	0.21284	0.88247	80	0.90628	0.3919	2.24747
21	0.67593	0.15542	1.06377	51	0.14936	0.08898	0.20762	81	0.70042	0.07232	0.36569
22	0.74151	0.22452	1.30322	52	0.20699	0.08973	0.23055	82	2.79361	0.31794	1.94605
23	0.80662	0.17215	1.67066	53	0.38957	0.27945	0.53767	83	0.4704	0.07633	0.37081
24	0.62751	0.05028	0.22199	54	0.24086	0.04457	0.20112	84	0.6243	0.07435	0.41588
25	0.19256	0.02943	0.10323	55	0.51241	0.34004	1.68994	85	0.07112	0.03609	0.10799
26	0.51801	0.04232	0.22614	56	1.56198	1.90934	8.7397	86	0.6294	0.03945	0.30174
27	0.52838	0.06053	0.27445	57	0.86337	0.16906	2.02475	87	0.50061	0.17798	0.73592
28	0.70014	0.17433	0.95367	58	0.29319	0.05206	0.25906	88	0.93252	0.22146	1.86277
29	0.75639	0.97257	4.04891	59	0.26706	0.06251	0.19045	89	0.18656	0.06746	0.15046
30	0.70411	0.4135	1.77721	60	0.20049	0.08754	0.22402				

B. Reproducibility and Submission Artifacts

Table VI lists every artifact needed to reproduce the reported numbers; the compliance declaration and the measured reproduction status follow it.

TABLE VI
Artifacts needed to reproduce the reported numbers.

Item	Value
Kaggle team	thisray
Primary candidate	D_union_H_G2
Kaggle submission	56353966 (Public 0.22610)
Prediction CSV	r15z_D_union_H_G2.csv on Kaggle, shipped as submission_D_union_H_G2.csv; 30,644 rows
CSV MD5	7a3fab85c4da99588f02d90f2e90c2b6
CSV SHA-256	9e9c254e6bfcc295e541ea9e2254ae93f94a53d7d
Network SHA-256	208c29160ebe9ecce8b9a83
Network SHA-256	925d43a38be18ba37f68e3ace927c7615a8ae2f73
Readout SHA-256	3c58013f62d5b53e36fff23
Readout SHA-256	78c7a1cde61d5ab38fcd630a2f165a83fd4e61f8
Second final	6334d42e8fc3a17be84609a
	D_union_H_G2_SC6, submission 56353958 (Public 0.22738); r15z_D_union_H_G2_SC6.csv on Kaggle, shipped as submission.csv
Its CSV MD5	5381dadd49140b23558cb18159088391
Its CSV SHA-256	dd01a43fd2db904f76f8b023adc00fe5e3db7241c
Code	b26f57feb59a1d34e10f8cf public release (Appendix D); numerical source commit 7eab1177
Configuration	pipeline_D_union_H_G2.json (all inference stages); base fine-tuning YAML released with the team report
Invocation	python inference.py --pipeline pipeline_D_union_H_G2.json --data-root <competition-root> --output submission_D_union_H_G2.csv, where the root holds test/ and index/test_windows.csv. The package default pipeline.json selects S-C6, so name the S-C3 pipeline explicitly
Environment	requirements.txt, 39 pinned versions; tested on GB10/T4; conda-linux-aarch64.lock for bit-identical reproduction on the reference architecture
Seeds	soup members 17; shared readout 17
Runs averaged	none; every reported score is one run of a deterministic pipeline

Compliance declaration.

1) Single model with one shared set of weights. Yes. One 43,669,432-parameter state dictionary (266 tensors), one 267-parameter shared readout, and one inference program process all 89 trajectories of all four platforms. The readout is a second-stage learner on the network output and the two physical proposals, fitted on released validation; it receives no platform label and holds no platform-specific parameters. The physics solvers hold no trained parameters, so there is no per-platform expert to dispatch to.

2) Platform classification, routing, or specialization at inference. Yes, among deterministic physics operators only, and strictly after the single weight set has run on

every trajectory. Fixed rules over the trajectory’s own raw IMU, and for the human rule its timestamps, select which operator if any refines it. The test split carries no platform column and the program raises if one appears; identifiers are read only to locate and order data, never as a predictive feature or as a key into a per-trajectory rule. Released platform labels were used in training as auxiliary supervision and to design the rules, and three gate constants were fitted on released data and frozen. One boundary we state explicitly: the ruling permitting hard routing describes a gate learned inside the network, whereas ours are fixed rules outside it. We do not claim it covers our case verbatim; every gate is described in full here and in the public release, and eligibility is the committee’s to judge.

3) Ensembling, checkpoint averaging, and TTA. Checkpoint averaging, yes: a uniform element-wise weight-space average of three members (six in S-C6), each its fixed terminal epoch-50 EMA (decay 0.999), all sharing one warm start; one averaged state dictionary is loaded and one network runs. Model ensembling, no: predictions of separate models are never combined. Test-time augmentation, yes, in temporal multi-view form: the one network is evaluated on every stride-one 16-window query view, so each window receives 1 to 16 passes (16 for interior windows), and its prior is the coordinate-wise median of those passes. No rotated, flipped, or noise-perturbed input is evaluated. The eight recurrent branches are averaged inside each forward pass and are not separate models. This TTA is unrelated to the abandoned physics test-time adaptation of Section III.

4) Test-set leakage, including model and checkpoint selection. No withheld label is an input to either selected final, and no weight, threshold, constant, or per-trajectory rule in them was fitted from withheld labels; all derive from released train and validation data, and no epoch was selected by any score. Development was nevertheless not independent of test feedback. The Public leaderboard and the official scoring service informed selection among finished candidates and several parent-pipeline choices: the detector union, the BODY operator variant, the rescale-then-smooth topology, the detector low-pass frequency, and the soup recipe pool. Separately, and more seriously, the team’s history contains an exploratory branch that submitted all-zero, scaled, and offset files, used differences in Public score to infer which trajectories are in the Public split, and uploaded three files built by per-trajectory selection over previously scored candidates, two of which also copied or fused predictions between test trajectories that appear to record the same flight with two sensors (Kaggle submissions 56324940, 56325119, 56325141). One of those still holds our best Public score; the branch was stopped and none of it was selected as a final. No output, label, constant, routing decision, or selected row from it occurs in either selected final, both of which are regenerated from raw IMU alone. Raw-IMU reproduction establishes what the

artifact computes; it does not adjudicate the earlier probing, which is the organizers’ call, and we have kept the timeline and evidence for it.

5) Development versus submitted system. Development executed some stages as separate programs over round-trip-safe CSVs; the release runs their pinned numerical routines in one process, replaces development paths with hash-checked manifests, removes evaluation code, and adds an offline guard. `pipeline_D_union_H_G2.json` reproduces `submission_D_union_H_G2.csv` byte for byte on the reference machine, with no numerical method, accepted gate, or output value changed. The package also carries a strict candidate and two output-gain pipelines for comparison. Neither selected final uses the additional fast-window gain stage. The gain worsened the Public score of the union candidate but improved that of the strict variant; neither gain variant was selected as a final. Reproduction status. On the reference machine (NVIDIA GB10, aarch64, the conda lockfile, network batch size 8) the package regenerates the CSVs of both selected finals byte for byte from raw IMU; this was verified in four independent clean-room runs of their pipeline family, the slowest taking 2,016 s. Two further runs of the unselected gain pipelines in the same batch were identical in all 91,932 float64 values but differ in text bytes, because the development files used a shorter decimal representation. The slowest run recorded anywhere for this package is 2,634 s on a heavily contended GPU, 37% of the two-hour budget. Determinism requires the pinned batch size, because a different batch size changes GPU kernel selection and the last bits. It also requires the full competition root for byte-exact comparison: changing the trajectory set changes network batch composition, and a five-trajectory root differed from the full 89 by 1.6×10^{-5} m/s. Context, proposal fits and aggregation remain within each trajectory; this small numerical difference is associated with batching, not cross-trajectory features.

On x86_64 with one NVIDIA Tesla T4 and `requirements.txt`, the primary candidate completed in 2,067 s using 489 MiB of GPU memory, with the same stage decisions as the reference run. The maximum absolute difference in a velocity component was 3.7×10^{-4} m/s. For S-C6 on the same T4, the tested PyTorch 2.13.0+cu130 and 2.10.0+cu128 stacks produced identical network priors; changing the tested CPU numerical packages changed final values by at most 5.6×10^{-9} . These comparisons locate most of the observed discrepancy in the network forward pass. The cross-environment outputs were compared numerically without rescaling on the official service. The pinned CUDA 13 wheel includes `sm_75` and newer targets. P100 and V100 require a compatible wheel; those GPUs were not tested.

Storage-frame diagnostics. Using released orientations, we compared gravity concentration before and after rotation into the world frame. This supports a body-stored family and a world-referenced family, with stored accel-

eration consistent with $R_{wb}(t)a_b(t)$ and stored angular velocity with $KR_{wb}(t)\omega_b(t)$, where $K = \text{diag}(1, -1, -1)$. This is our diagnostic interpretation of the released streams, not an independently verified account of their production. A constant sensor remount is absorbed by that storage law when the corresponding extrinsics are updated; applying the same constant rotation to the stored acceleration and gyro is generally not a physical remount. No test orientation is read by the pipeline.

C. Final Predictions

The primary predictions ship in the public release as `submission_D_union_H_G2.csv` and the S-C6 predictions as `submission.csv`, each byte-identical to the file uploaded to Kaggle (`r15z_D_union_H_G2.csv` and `r15z_D_union_H_G2_SC6.csv`, respectively). The format is `window_id,vx,vy,vz`, one row per test window, 30,644 rows plus the header, with `window_id` taken from `index/test_windows.csv`.

D. Model and Inference Code

The public release is <https://huggingface.co/gn016/97933/tartanimu-iros2026-unified-model>. It contains both selected weight sets with frozen manifests, pinned requirements, the single inference entry point, every pipeline file, and the CSVs of both selected finals. The release refuses to run if a weight file’s SHA-256 is unexpected, and disables network access before importing numerical libraries. It also carries the organizers’ starter-template team report, which names the AI systems behind the workflow disclosed in the acknowledgment, and the base fine-tuning configuration file.

E. Negative Results Ledger

Table VII lists the abandoned ideas of Section III with the measurement that closed each one. Entries are ordered by how much they changed our development strategy, not by how much score they lost. Where a receipt did not retain a cost we say so rather than estimate it.

TABLE VII

Ideas we tried and abandoned. Validation experiments use the released validation split; subsets and platform-specific measurements are identified in the entries. Official figures use the all-89 scoring service. Positive score changes are worse.

Idea	Outcome	Cost
Geometric median over the eight branches	validation +0.014760, $30\times$ the preregistered +0.0005 rejection bound; AVE +5.5%, ATE20 +10.0%	re-aggregation only
Per-trajectory rotor-drag coefficient	identified on 48/52; substituting the identified values +0.000153, free coefficient +0.000167, global median -0.000047 , label-picked grid improvement -0.00066	oracle analysis, no training
Physics test-time adaptation	11/16 validation trajectories worsened; one human trajectory’s AVE $0.1182 \rightarrow 13.1750$ m/s, speed $59.6\times$ prior (1,029 windows); guard passed 3 harmful, blocked 2 beneficial cases	1,291 s of fits
Rotation augmentation as an invariance fix	car proxy gap -93.3% (exposure-matched, post-freeze); the same recipe moved official all-89 by -0.00139 , below the preregistered 0.002 adoption threshold	3 fine-tunes; 1 wall clock not retained
Direction-targeted loss term	drone per-window direction error 20.433° moved by 0.054° against a 1.5° requirement	149 min GB10
Averaging five independently trained priors	$20.433^\circ \rightarrow 20.517^\circ$, worse than the best single member (20.285°)	inference on existing arms
Lowering the BODY speed floor from 2 to 1 m/s	BODY-trajectory AVE +0.0299 m/s, worst trajectory +0.2532 m/s, validation score +0.0014; failed every preregistered criterion	analysis only
Fast-window gain constant	validation LOO -0.00103 , bootstrap p95 -0.00093 ; Public $0.22738 \rightarrow 0.22815$	not recorded
Task-vector composite fine-tune	validation -0.0005 against a pre-set -0.0010 threshold	not recorded
Time rescaling of fast windows	implied speeds $19.0/18.2$ m/s against a released maximum of 10.35 m/s; +0.018 to +0.020	analysis only
Privileged-label supervision probe	-0.000169 and $+0.000082$ against a -0.002 requirement	not recorded

F. Inference Constants

Table VIII lists the fixed constants of the stages after the network, each with its value and the data it was chosen on. Statistic names follow the released code (tartanimu_release/detectors.py and the scripts it calls). No constant was fitted on test labels; the choices that test feedback informed are the ones named in answer 4 of Appendix B, and they are marked below.

TABLE VIII

Fixed constants of the inference pipeline and the data each was chosen on. Train and val are the released splits; frame-family and platform labels are released labels.

Stage	Constants and values	Chosen on
Readout	267 learned parameters (Section II-E)	val (80 trajectories), train-only-twin prior, AVE loss
BODY gate	$\text{rot_ratio} \leq 1.2$, $\text{gait} < 0.35$, $ \text{corr_body} \leq 0.45$, mean tilt $\geq 3^\circ$; fires if any of the 0.5, 1, or 5 Hz low-pass versions fires	thresholds: train/val with frame-family labels; three-frequency union: test feedback
BODY operator	rotor-drag coefficient 0.30; prior rescaled by the ratio when the network-free mean speed is ≥ 2 m/s and $\geq 1.4\times$ the prior's; zero-velocity samples where the 2 s rolling std of $ \omega $ is below 0.02 rad/s	coefficient: released body-stored drones; rescale bounds: nine val body-stored drones with a train-only prior and a halved-prior stress case; operator form and prior weighting: test feedback
WORLD gate	$\text{corr_body} \leq -0.45$, mean tilt $\leq 3.5^\circ$, $\text{rot_ratio} \leq 1.6$, hf50 ≤ 0.10 ; same 0.5/1/5 Hz union	thresholds: train/val with frame-family labels; union and 1 Hz family low-pass: test feedback
WORLD G1	fused branch reverts if the fit's median residual exceeds 0.657 m/s or its mean change exceeds 0.193 m/s; switches to pure physics when the spectral misfit ratio exceeds 0.1198 (correlation $\rho = 0.5$)	residual bounds: released WORLD flights; 0.1198: 39 val WORLD flights under stress-test priors
WORLD G2	20-window segments; 0.5/0.5 blend with G1; accept if heading uniqueness ≥ 5 , speed ratio ≤ 1 , 90th-percentile attitude misfit $\leq 2.5^\circ$, and at least 10 windows; attitude-fit drag constant 0.39	train only, with synthetic priors; selector thresholds from a mechanical grid search
Ground gate	mean tilt $< 30^\circ$, duration > 60.5 s, $\text{rot_ratio} \geq 1.43$; dog if $22.6885 \text{still_frac} + 11.4416 \ln(\text{std } a) - 9.6867 > 0$, otherwise car; 5 Hz low-pass	thresholds: train/val; the three coefficients fitted on released car and dog trajectories with platform labels
Dog and human operator	30 s segments; zero-velocity samples where the 2 s rolling std of $ \omega $ is below 0.02 rad/s; 0.5/0.5 blend with the incoming prior	released ground trajectories; reuse for human selected by a preregistered test on val human trajectories
Car operator	60 s segments; zero-velocity samples where the 2 s rolling stds of $ \omega $ and $ a $ are below 0.02 rad/s and 0.2 m/s ² ; no outer blend	released ground trajectories
Human gate	mean tilt $> 60^\circ$ and duration > 300 s	earlier development rule, re-checked on all 475 released trajectories
Guards	BODY, dog, car, and human revert on non-finite output or any window above 30 m/s	fixed safety bound
Soup	uniform member weights; fewer members preferred within 0.0001	preregistered rule on val with train-only twins
Fast-window gain	$\times 1.0200$ or $\times 1.0225$ on windows above 1 m/s	val; used by neither selected final

Gate statistics. Mean tilt is the angle between the trajectory's mean specific force and the sensor z axis. hf50 is the fraction of accelerometer power at or above 50 Hz, and still_frac the fraction of samples whose 2 s rolling standard deviation of $|\omega|$ is below 0.02 rad/s. gait is the power of $|a|$ between 0.8 and 4 Hz divided by its power between 0.1 and 100 Hz. rot_ratio and corr_body compare the time derivative of the low-passed specific force f with $-\omega \times f$, the change that rotation alone would produce, both taken perpendicular to f : rot_ratio is the ratio of their RMS magnitudes and corr_body their normalized correlation. Low-pass filters are zero-phase fourth-order Butterworth filters at the stated frequency. Duration is the sample count divided by 200 Hz in the ground gate and the span of the timestamps in the human gate.